## KOVALENKO, 34, M

2026-08-29 14:32 · Stab. point *Sokil*

## WHAT THE PATIENT SAID — verbatim, no clinician inference

| Point | Time | Said | Requests |
|---|---|---|---|
| POI | 11:05 | “head hurts, can’t see right” | water |
| CCP | 12:40 | neck pain turning left | water ×2 |
| Stab. Sokil | 14:20 | **headache worse lying flat** | asks about brother ×2 |

**Note.** “Worse lying flat” is positional and is evidence about intracranial pressure that the model did not produce. It arrives from the patient and is not reconciled with the model output below — where the two disagree, both are shown.

### VITALS (14:28 — 4 min old)

| | |
|---|---|
| BP | 104/62 (MAP 76) |
| HR | 118 |
| $SpO_2$ | 94% RA |
| RR | 24 |
| GCS | 14 (E4 V4 M6) |
| Temp | 36.4 °C |
| Cap refill | 3 s |

### MODEL INPUT

| | |
|---|---|
| Recording | 081, 100 Hz ICP + ABP |
| Manoeuvre | jugular compression, 118 s |
| Window | −120 to +268 s |
| Fit | RMSE 0.31 mmHg / 261 beats |
| Identifiable | 13 of 16 windows |

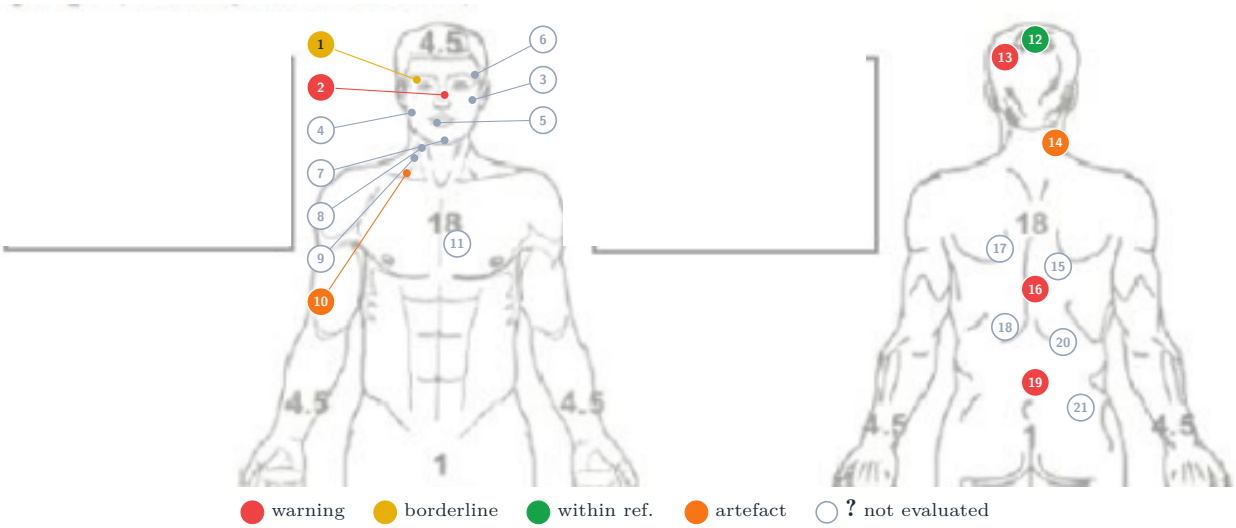
RMSE is the typical gap between the model’s ICP and the measured ICP. 0.31 mmHg is small; it means the model reproduces the pressure, not that the parts below are right.

# PULSATION ANALYSIS

the CSF and venous circuit as 9 compartments and 12 interfaces · colour = status, number = key below

## The digital twin suggests investigating these zones.

Start with **red**. **Orange** is the model correcting its own arithmetic — not a finding in this patient. Grey was not measured at all.




### HOW TO READ THIS — four terms on this card that do not mean what they usually mean

**w 0.63** — how far *outside* the normal band that part sits. 0 = at the edge of normal, 1.0 = at the danger bound, higher = further out. It is **not** a probability and **not** a severity score.

**?** — nothing on this card measured that part. It means **not measured**. It does **not** mean normal — 13 of the 21 parts are like this.

**artefact** — the model bent this number to make its own arithmetic work. It is a statement about the model, **not about this patient**. Do not read the value.

**the dots** — lumped parts of the CSF and venous circuit, drawn where that region roughly is. They are **not surgical landmarks**.

**Where the numbers come from.** A two-minute jugular compression, held by hand. Before it starts almost nothing here can be measured at all; the compression is what makes the venous numbers estimable, and **Identifiable** above counts the windows where that worked. Where it did not, the value is a guess and is marked **?**.

### COMPONENT KEY

| | | | | | |
|---|---|---|---|---|---|
| 1 Cerebral arterial | w 0.39 | 8 Sinus to jugular bulb | ? | 15 Vertebral plexus | ? |
| 2 Intracranial CSF | **w 0.63** | 9 Jugular bulb | ? | 16 Spinal dural sac | **w 0.63** |
| 3 Cerebral venous | ? | 10 Internal jugular | **ARTEFACT** | 17 Azygos return | ? |
| 4 Arteriolar bed | ? | 11 Systemic venous | ? | 18 Paravertebral | ? |
| 5 Capillary bed | ? | 12 Dural sinus | w 0.11 | 19 CSF absorption, spinal | **w 1.53** |
| 6 Cerebral venous outflow | ? | 13 CSF absorption, cranial | **w 1.53** | 20 Plexus drainage | ? |
| 7 Craniospinal CSF channel | ? | 14 Vertebral collateral | **ARTEFACT** | 21 Plexus shunt | ? |

**13 of 21** components carry **?**: no streamed parameter governs them. **?** means **not estimated**, never **normal**. The two **artefact** components are rulebook **R1** (tube-law escape) — the fit inflated the resting jugular to 6.12 $cm^2$ against a physiological 0.8–2.0 and blocked the collateral to manufacture the measured ICP step. Neither is a statement about this patient.

**Provenance.** Component weights, values and the R1 determination are **real output** for recording 081, final window ($t_{end}$ = 268.4 s). Name, age, sex, vitals, evacuation points and transcript are **synthetic** — the dataset contains none of these. **Weight** is the excursion beyond the healthy band (1.0 at the pathological bound), discounted by identifiability; it is not a probability. **No healthy controls exist in this dataset** — every bound is from literature. The underlying model overestimates ICP pulse amplitude under compression by ∼7× from a diagnosed cause, which is why the jugular carries a QC bound rather than a clinical one. Pin positions are schematic — the anatomical region a lumped element stands for, not a surgical landmark. Not a diagnosis. Confirm before acting.